AI for Teaching, Learning and Collaborating

D-Lab AI Pulse Workshop

Materials for all Pulse Workshops available here:
https://github.com/dlab-berkeley/AI_Pulse

The Bottleneck: Learning

The traditional model: One instructor, dozens of students, teaching "by the average."

What this looks like in practice:

- A student is stuck on a concept. office hours are five days away
- Course evaluations: half say "too fast," the other half say "too slow"
- A grad student spends weeks in a project, only to discover the idea isn't novel

Material is presented at one pace, in one way. Students who need more time wait. Students who need less time disengage.

The Bottleneck: Teaching & Collaborating

Teaching:

- GSIs answer the same email 30 times a semester: "Where is the rubric?" "Can I submit late?" "How do I request DSP accommodations?"
- Instructors have ideas for improving the syllabus, but postpone it due to the time required to prepare material
- Creating new exams means manually modifying old problems

Collaborating:

- Co-authors have results in one Google Doc, a draft in another, data files in email attachments, lit review notes in Slack, ideas/tasks discussed in meetings are lost
- Organizing and keeping track of progress takes time away from research

What is NotebookLM?

Google's source-grounded AI powered by Gemini

Key differentiator: Answers come from selected documents, not the internet.

How it works:

- Upload: PDFs, Google Docs/Slides, URLs, YouTube links, audio files, copy-paste text
- Every claim has inline citations back to the source material
- Can search the web and add new sources to the notebook

Sharing: You can share notebooks. viewers interact with the AI but cannot edit sources or see your instructions

Free: https://notebooklm.google.com/

Today's Roadmap

	Section	What we'll cover
1	Learning	Learning from course materials; exploring a literature
2	Teaching	The Bureaucratic Agent; exam creation & study tools; incorporating material into slides; "Conversation with Weber"
3	Collaborating	Technical knowledge bases; group brainstorming; pre-paper organizer

Learning

with NotebookLM

Demo 1: Learning from Course Materials

Setup: Upload lecture slides from a course (+ YouTube lecture links, Google Docs, URLs)

One set of course materials → six ways to engage:

1. Chat with sources. ask questions, get cited answers
2. Audio overview. podcast-style (Deep Dive, Brief, Critique, Debate)
3. Study guide. organized by topic
4. Flashcards & quizzes. with "Explain" button for wrong answers
5. Mind map. concept map across uploaded slides
6. Learning Guide mode. Socratic tutoring, adapts to student level

Takeaway: The student chooses the modality that works for them.

Try it: https://notebooklm.google.com/notebook/7dc90650-6279-4cc5-a32b-1084d4b672c6

Demo 2: Exploring a Literature

Andrej Karpathy
@karpathy

I'm starting to get into a habit of reading everything (blogs, articles, book chapters,...) with LLMs. Usually pass 1 is manual, then pass 2 "explain/summarize", pass 3 Q&A. I usually end up with a better/deeper understanding than if I moved on. Growing to among top use cases.

On the flip side, if you're a writer trying to explain/communicate something, we may increasingly see less of a mindset of "I'm writing this for another human" and more "I'm writing this for an LLM". Because once an LLM "gets it", it can then target, personalize and serve the idea to its user.

4:29 PM · Nov 17, 2025 · **2.8M** Views

Andrej Karpathy on reading with LLMs: pass 1 manual, pass 2 explain/summarize, pass 3 Q&A.

Try it: https://notebooklm.google.com/notebook/53ebc479-f5b2-4db6-bba3-5cb95afbcb60

Setup: 5–6 papers uploaded on a research topic.

1. Cross-paper Q&A. "What are the main methodological approaches?" Cited answers from multiple papers.
2. Synthesis. "Where do these papers agree and disagree?"
3. Novelty check. "What has already been done and what gaps remain?"
4. Search for new sources. find and add relevant papers.

Takeaway: Figure out in an afternoon what used to take weeks.

Tools to Keep an Eye On: Learning

Tool	What it does	Link
Khanmigo	Khan Academy's AI tutor. Socratic method. guides you to the answer instead of giving it. Covers math, science, coding, humanities.	khanmigo.ai
Microsoft Study	Microsoft's AI learning agent. Adaptive exercises, flashcards, and quizzes from your materials. Free 12-month access for higher ed students.	microsoft.com
SciSpace	Deep single-paper reading. Paragraph-level Q&A, in-line explanations, citation export.	scispace.com

Teaching

with NotebookLM

Demo 3: The Bureaucratic Agent

Setup: Upload course syllabus + UC Berkeley code of conduct, DSP rules, grading policies, late submission policy.

Student questions (pre-generated):

- "When is the midterm and what does it cover?"
- "Can I submit this assignment late? What's the penalty?"
- "How do I request DSP accommodations?"
- "What counts as academic dishonesty on the group project?"

Each answer cites the specific policy document and section.

Sharing: Students interact with the AI but cannot see your instructions or edit sources.

Also works for: Undergraduate advising, lab onboarding, department admin.

Try it: https://notebooklm.google.com/notebook/175d8c57-d907-44be-bbdd-da1058254aec

Demo 4a: Exam Creation (Instructor Side)

Setup: A notebook loaded with 3–4 previous exams/problem sets.

Demo sequence:

1. Generate new problems. "Create 10 new problems based on the style and difficulty of the previous exams. Modify the numbers, names, and prose, but keep the underlying concepts." The instructor picks 5 for the new exam.
2. Generate a solutions package. "Create detailed solutions for problems 2, 4, 6, 8, and 10."
3. Fix logistics. "Update the header to say 'Midterm 2, Spring 2026' and change all dates."

Takeaway: The instructor's job shifts from typing assignments to planning them.

Try it: https://notebooklm.google.com/notebook/d28c0e9c-09ea-4479-b828-c01d03ea209f

Demo 4b: Student Study Tool (Student Side)

Setup: Same notebook (previous exams), configured for student use.

Custom instructions to NotebookLM:

> *"Students will ask you practice questions. Do NOT give them the answer directly. Instead, give hints and ask guiding questions to help them get unstuck. Only reveal the solution after the student has made a genuine attempt."*

Student interaction: "I'm stuck on problem 3 from the 2024 midterm." $\rightarrow$ NotebookLM responds with a hint, not the answer. Guides the student through.

Takeaway: The same material serves two purposes. exam creation for the instructor, Socratic tutoring for students. by changing the instructions on the shared notebook.

Try it: https://notebooklm.google.com/notebook/d28c0e9c-09ea-4479-b828-c01d03ea209f

Demo 5: Incorporating New Material

Setup: Existing lecture slides + 2–3 new papers the instructor wants to integrate.

Prompt:

> *"I want to incorporate the findings from [new paper] into my Week 6 lecture on [topic]. Suggest where in the existing slides this material fits, and draft 2–3 new bullet points or a new slide."*

NotebookLM identifies the right place in the existing slides and drafts new content grounded in both the paper and the existing material.

New: NotebookLM now supports prompt-based revisions. you can ask it to rewrite, reorganize, or refine its outputs iteratively, right in the chat.

Takeaway: Course maintenance. updating slides semester after semester. becomes a conversation rather than a manual rewrite.

Try it: https://notebooklm.google.com/notebook/2e5cc766-57e4-414f-bbd6-241bf9b5678c

Demo 6: "Conversation with Max Weber"

Setup: Upload Weber's major works. Configure instructions:

> *"You are a scholar deeply knowledgeable about Max Weber's work. Answer questions from Weber's perspective, grounding your responses in the uploaded texts. Engage Socratically."*

A student asks: "Are wealth taxes justifiable?"

- NotebookLM responds from Weber's perspective, citing his writings on bureaucracy, legitimate authority, and economy and society
- The student pushes back $\rightarrow$ dialectical engagement, still grounded in Weber

The pedagogical point: A hammer becomes much more interesting if one is already holding a nail. The conversation creates the motivation to read the primary texts.

Try it: https://notebooklm.google.com/notebook/42a25599-d4f9-47bb-8e09-d7a2d6585bb2

Tools to Keep an Eye On: Teaching

Tool	What it does	Link
Mizou	Build custom AI tutoring chatbots grounded in your course material. Designed specifically for education.	mizou.com
Curipod	AI-powered interactive lesson builder. polls, word clouds, open-ended prompts for real-time engagement.	curipod.com
Quizlet AI	AI-enhanced flashcards and study tools with spaced repetition.	quizlet.com

Collaborating

with NotebookLM

Demo 7: Technical Knowledge Base

Setup: Upload EML cluster documentation, SLURM docs, and example job scripts.

Demo sequence:

1. Practical question. "How do I parallelize this job across 8 nodes? Here's my current script." NotebookLM suggests changes grounded in the actual cluster docs.
2. Troubleshooting. "My job keeps getting killed after 2 hours. What's the memory limit?" Pulls the relevant policy from the docs.

Also works for:

- Lab shared infrastructure: "How do I use the confocal microscope after hours?"
- Shared codebases: upload documentation and README files
- Grant admin: upload NSF/NIH guidelines → "Can I charge travel to this grant?"

Takeaway: Turn the manual into a conversation.

Try it: https://notebooklm.google.com/notebook/dd8b8156-5c97-4c16-ad8e-10f3567da3de

Demo 8: Group Brainstorming & Project Development

Setup: A shared notebook. team members have each added papers, data descriptions, and pasted notes.

Demo sequence:

1. The "Add Note" feature. team members add ideas and observations as notes. Each person's contributions become part of the shared knowledge base.
2. Search for new material. "Find recent papers on [method we're considering]." New sources get added.
3. Generate a project report. "Based on everything in this notebook, generate a feasibility report: (a) what we know, (b) where we agree, (c) open questions, (d) what we still need to figure out."

Takeaway: The notebook becomes a living project workspace. One place where all the thinking accumulates and can be queried.

Try it: https://notebooklm.google.com/notebook/b6c22a24-b2fb-4f63-92e3-cd6577bbd54f

Demo 9: Pre-Paper Organizer

Setup: Collaborators have each added their piece. empirical results, theoretical model, literature review notes and citations.

Demo sequence:

1. Draft a paper structure. "Draft a detailed outline. Identify the main argument, suggest how theory connects to empirics, flag gaps."
2. Identify gaps. "Where does the theoretical framework make predictions that the empirical work doesn't test?"
3. Draft an abstract. grounded in the actual results and theory uploaded, not generic filler.

Takeaway: A big obstacle in a multi-author paper is agreeing on the structure and synthetizing work. NotebookLM provides a first draft based on what everyone has actually contributed.

Try it: https://notebooklm.google.com/notebook/0b8e1f86-eae5-4a5a-a300-9c9433e1cfad

Tools to Keep an Eye On: Collaborating

Tool	What it does	Link
Mural	AI-assisted visual collaboration. digital whiteboards for brainstorming, affinity diagrams, research synthesis.	mural.co
Notion AI	Workspace with AI writing, meeting summaries → action items, and calendar integration. Strong for project management.	notion.so
Otter AI	Meeting transcription with speaker identification, cross-meeting search, and mid-meeting Q&A. Free tier: 300 min/month.	otter.ai

What We Covered

Application	What NotebookLM does
Learn from course materials	Chat, audio overviews, flashcards, quizzes, mind maps, Socratic tutoring
Explore a literature	Cross-paper Q&A, synthesis, novelty checks, source discovery
The Bureaucratic Agent	Syllabus/policy Q&A shared with students
Exam creation & study tools	Generate new problems, solutions, and Socratic study tools
Incorporate new material	Integrate new readings into existing slides
"Conversation with Weber"	Socratic dialogue grounded in primary sources
Technical knowledge base	Turn documentation into conversational Q&A
Group brainstorming	Shared notebook with notes, search, feasibility reports
Pre-paper organizer	Draft paper structure from team contributions

Discussion

1. Is struggling the point? If AI explains any concept instantly, do students lose something? Is working through difficulty how we actually learn. or is it a bottleneck we've romanticized?
2. "Conversation with Weber". help or hurt? Does talking to an AI version of a thinker prime students to engage with primary sources, or let them skip the reading?
3. The instructor's new role. If AI handles tutoring, Q&A, material generation, and exam creation. what's left? Is what's left the most important part?
4. The exam question. AI humanizer tools already rephrase text to evade detection, and AI gadgets are becoming thumb-sized. Should we try to shield students from AI on assignments. or redesign assessments to embrace it? (See: companion.ai/einstein)
5. Who benefits? A first-generation student now has a personalized tutor. But students with more tech fluency adopt faster. Narrow or widen the gap?

Resources

Primary tool: NotebookLM. https://notebooklm.google.com/

Tools to keep an eye on: Khanmigo (khanmigo.ai), Microsoft Study (microsoft.com), SciSpace (scispace.com), Mizou (mizou.com), Curipod (curipod.com), Quizlet (quizlet.com), Mural (mural.co), Notion AI (notion.so), Otter AI (otter.ai)

Demo notebooks (click to explore):

- Demos 1–2 (Learning): Course Materials — Literature
- Demos 3–6 (Teaching): Bureaucratic Agent — Exams — New Material — Weber
- Demos 7–9 (Collaborating): Knowledge Base — Brainstorming — Pre-Paper

D-Lab: https://dlab.berkeley.edu/ All materials: https://github.com/dlab-berkeley/AI_Pulse